

VILO ACUNA INTL, Cuba

WMO: 783334

Lat: 21.616N

Lon: 81.546W

Elev: 3

StdP: 101.29

Time Zone: -5.00 (CUB)

Period: 10-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.9	19.9	10.9	8.1	20.4	12.1	8.8	21.2	10.8	22.5	10.3	23.1	5.8	20	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.8	32.2	25.9	32.1	25.8	31.9	25.7	27.3	30.4	27.0	30.2	26.7	30.0	4.2	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.2	21.7	29.3	26.1	21.5	29.3	25.9	21.3	29.3	86.8	31.0	85.3	30.4	84.0	30.2	29.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.8	9.0	8.3	16.8	33.3	1.8	0.9	15.4	33.9	14.4	34.4	13.3	34.9	12.0	35.5		
(5)			12.9	28.0	1.6	0.9	11.7	28.6	10.8	29.1	9.8	29.6	8.7	30.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.0	24.1	24.8	25.2	26.8	27.6	28.7	29.3	29.3	29.1	28.0	26.2	25.1
	DBStd	2.29	1.84	1.68	1.49	0.93	1.13	1.05	0.97	1.12	1.13	1.32	1.48	2.06
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	1	0	0	0	0	0	0	0	0	0	0	0	1
	CDD10.0	6219	438	416	472	504	546	561	597	599	573	558	486	469
	CDD18.3	3178	180	182	214	254	287	311	339	341	323	300	236	212
	CDH23.3	33377	1008	1207	1523	2440	3138	3873	4479	4533	4188	3433	2001	1553
	CDH26.7	11114	87	136	213	527	912	1546	2036	2104	1839	1163	357	195
Wind	WSAvg	4.4	5.0	4.7	5.0	4.5	4.4	4.2	3.8	3.9	3.5	4.5	5.1	4.7
Precipitation	PrecAvg	1140	41	30	31	50	123	159	112	148	175	158	68	44
	PrecMax	1766	125	97	88	161	404	390	204	213	401	337	233	187
	PrecMin	701	12	4	12	17	27	58	59	90	82	44	1	5
	PrecStd	194	28	20	16	30	85	75	35	31	61	82	50	42
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.0	29.2	29.2	30.2	31.2	32.2	33.1	33.1	34.0	31.9	30.2	29.2
		MCWB	23.6	23.8	23.4	24.6	25.1	25.8	26.4	26.4	28.1	25.9	25.0	24.5
	2%	DB	28.1	28.8	29.1	30.1	31.0	32.0	32.2	32.2	32.2	31.1	30.0	29.0
		MCWB	23.2	23.5	23.4	24.7	25.0	25.6	25.8	25.8	26.1	25.9	25.0	24.3
	5%	DB	27.8	28.1	28.2	29.2	30.2	31.2	32.0	32.1	32.0	30.9	29.1	28.2
		MCWB	23.1	23.0	22.9	24.1	24.5	25.6	25.7	25.7	25.8	25.9	24.6	23.6
	10%	DB	27.0	27.8	28.0	29.0	30.0	31.0	31.8	31.8	31.2	30.2	28.8	27.9
		MCWB	22.6	22.9	22.7	23.9	24.4	25.6	25.7	25.7	25.8	25.5	24.4	23.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.7	24.8	25.1	25.8	26.6	27.6	27.7	27.7	28.1	27.2	26.1	25.7
		MCDB	27.2	27.8	28.2	29.3	29.5	30.1	29.7	30.4	32.5	29.7	28.4	27.7
	2%	WB	24.1	24.2	24.4	25.3	26.1	27.0	27.1	27.1	27.2	26.8	25.7	25.1
		MCDB	26.5	27.1	27.6	28.6	28.9	29.9	30.3	30.6	30.7	29.6	28.2	27.5
	5%	WB	23.6	23.7	23.9	25.0	25.7	26.6	26.7	26.7	26.7	26.6	25.3	24.6
		MCDB	26.3	26.7	27.2	28.2	28.6	29.6	30.2	30.4	30.1	29.4	28.1	27.1
	10%	WB	23.2	23.4	23.4	24.6	25.2	26.3	26.5	26.5	26.5	26.2	24.9	24.2
		MCDB	26.1	26.4	26.8	27.8	28.3	29.4	30.1	30.2	30.0	29.2	27.9	26.8
Mean Daily Temperature Range	5% DB	MDBR	4.5	4.7	4.8	4.6	4.4	4.4	4.6	4.8	4.7	4.1	3.9	4.1
		MCDBR	4.8	5.0	5.1	4.8	4.7	4.7	5.0	4.9	5.0	4.5	4.1	4.2
		MCWBR	2.6	2.4	2.5	2.2	2.3	2.3	2.7	2.5	2.3	2.2	2.2	2.2
	5% WB	MCDBR	4.6	4.8	4.8	4.7	4.5	4.4	4.7	4.8	4.6	4.3	3.8	4.3
		MCWBR	2.7	2.5	2.5	2.3	2.5	2.5	2.8	2.6	2.5	2.3	2.2	2.4
Clear-Sky Solar Irradiance	taub	0.366	0.371	0.378	0.409	0.437	0.467	0.474	0.445	0.418	0.401	0.379	0.376	
	taud	2.569	2.552	2.521	2.420	2.397	2.343	2.313	2.422	2.497	2.545	2.555	2.566	
	Ebn at Noon	907	923	927	896	862	829	824	852	876	880	886	883	
	Edh at Noon	93	100	107	120	121	127	131	118	108	99	93	91	
All-Sky Solar Radiation	RadAvg	4.37	5.26	6.14	6.77	6.46	6.09	6.58	6.29	5.59	4.93	4.46	4.08	
	RadStd	0.30	0.20	0.25	0.16	0.67	0.38	0.39	0.25	0.36	0.43	0.23	0.23	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page